Rohit B

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.19
The Hindu Senior Secondary School	Class XII (Central Board of Secondary Education)	2021	94.80%
The Hindu Senior Secondary School	Class X (Central Board of Secondary Education)	2019	88.80%

WORK EXPERIENCE

Ramco Cements | IT Infrastructure & Network Intern

- Traced end-to-end traffic through a live **enterprise data centre**, from **core switches** through **distribution layers** out to **edge ports**, with redundant uplinks and dual power feeds removing single points of failure.
- Designed a **3-branch network** with per-floor segmentation and differing user counts, applying **CIDR subnetting**, **classless addressing** and **DHCP DORA** lease allocation across each site.
- Examined the perimeter framework of **firewalls**, **IPS / IDS**, **VPN tunnelling** and **patch management**, and studied how **Layer-3 switches** handle **inter-VLAN routing** without dedicated routers.
- Investigated link failures using standard connectivity diagnostics, and mapped how internet access is distributed from the head office out to each branch, including the **OSI layers** involved at every hop.

Tech Mahindra | AI / ML Intern

- Developed **ChatResume**, a **CV screening assistant** that reads a folder of documents and answers questions about the candidates, using **PDF extraction**, **vector search** and an **LLM** for grounded answers.
- Diagnosed each CV being stored as a **single vector** against a **256-token limit**, cutting off **80%** of its text, then improved result quality from **0.082 to 0.732** across **2,483 real CVs**.
- Engineered **CyberIntelBot**, a **security log triage assistant** that parses **Windows**, **syslog**, **JSON** and **CSV** formats and matches behaviour against **607 MITRE ATT&CK techniques**, citing the source line as evidence.
- Validated it on **18,000 lines** of real third-party logs spanning **nine separate systems**, tuning the alert rate down to **0.17 per 1,000 lines** while still catching genuine brute force attempts.

PROJECTS

Minigit | Git Re-implementation with Semantic Merge | Python | GitHub

- Architected a **Git re-implementation** in **Python** matching its **on-disk object format** byte for byte, so native **git log**, **status** and **fsck** run unchanged against any repository it creates.
- Verified correctness by **differential comparison** against real Git over randomly generated command sequences, with all **200 of 200** runs producing byte-identical object stores on both sides.
- Blocked a merge that **real Git approves** and the program cannot survive, one branch renaming a function while another still calls the old name, by comparing **syntax trees** rather than lines of text.

Verifiable URL Shortener | Python, FastAPI, PostgreSQL, Docker | GitHub

- Deployed a link shortener keeping a **tamper-evident record** of every destination change, verified across all **8,385** history sizes, so anyone can check whether a link was quietly repointed.
- Accelerated the redirect path **8,710 times**, from **67.8 ms to 7.78 microseconds**, by batching click counts off the request path and reusing **pooled database connections** instead of opening one each time.
- Eliminated unsafe destinations by **tracing the full redirect chain** and judging the final hop rather than the address submitted, with **threat feeds** checked at every step along the way.

RVSIM | RISC-V Emulator with Pipeline and Cache Models | C++ | GitHub

- Coded a **RISC-V processor simulator** in **C++** that executes assembly programs and reports the **cycle cost** of each

one, including a **five-stage pipeline** model and a configurable **cache**.
- Demonstrated that two programs executing **identical instruction counts** differ by **1.88 times** in total cycles, purely from the order in which they walk memory, row-major against column-major.
- Classified every cache miss as **compulsory, capacity or conflict** with a **fully-associative shadow cache**, showing the standard remedy bought nothing while **capacity** cut cycles to **175,136**.

SKILLS

- **Languages:** Python, C++, SQL
- **CS Fundamentals:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Computer Architecture
- **ML:** scikit-learn, TensorFlow, Keras, GANs, model calibration
- **Backend:** FastAPI, REST APIs, PostgreSQL, Docker, Git
- **Security & Networking:** DNS, DNSSEC, BIND, MITRE ATT&CK, firewalls, Wireshark, IPS / IDS, VLAN, CIDR, DHCP
- **NLP & Retrieval:** RAG, BM25, Pinecone, Sentence-Transformers

POSITIONS OF RESPONSIBILITY

Teaching Assistant, Vigil Lab | Department of CSE, IIT Hyderabad